A

Project Report On

SmartCart: Transforming Education and Inclusion through Innovation

Table of Contents

1. Introduction....2
2. Requirement Design....4
3. Methodology/Design....9
4. Implementation and development....15
5. Testing....20
6. Results and Analysis....24
7. Challenges and Limitations....30
8. Future Work....33
9. Conclusion....35
10. References....37

1. Introduction

Background and Motivation

Time represents a crucial element in modern life because long waits at supermarket billing stations deliver continuous frustration to customers. The standard shopping process presses consumers to wait in check-out lines manually thus creating customer dissatisfaction and increased waiting frustration. These problematic procedures create two major impacts by causing customer dissatisfaction and adding unnecessary work for staff members.

Because of its RFID component and automated payment, the SmartCart addresses this problem in regular shopping carts. It assists in achieving the UN Sustainable Development Goals by making retail more efficient and convenient for customers. For Goal 8: Decent Work and Economic Growth, companies can help by changing high-workload jobs to allow employees to work with more technology. Under goal 9, industry, innovation and infrastructure, advancements are being made in retail systems that streamline how companies operate. Ensuring that billing and inventory are accurate in the retail process address Goal 12: Responsible Consumption and Production. The adoption of technology and thoughtful design helps SmartCart improve the efficiency of processes.

Problem Statement

Most supermarkets and retail stores use manual billing systems that create excessive lines, prolonged waiting periods and resource waste and degrade shopper satisfaction. Consumers typically stay longer at checkout stations as opposed to their time spent shopping for items. Poor customer satisfaction together with customer abandonment occurs during peak times and festive periods.

An intelligent automated system needs development to enable customers perform self-service shopping and billing without waiting in lines at checkout areas. The SmartCart aims to resolve the issue through product recognition technology based on RFID combined with self-billing features built within the shopping cart.

Objectives

The main business goals of the SmartCart project focus on:

1. The goal is to build an RFID-based shopping cart system which identifies products being bought during actual transactions.
2. The system integrates a display system which calculates and displays the total cost for automation of billing operations.
3. Customers will have a convenient shopping journey because the system performs automatic billing without requiring human assistance or waiting at queues.
4. The integration between hardware components and program software will show their collective ability to boost user satisfaction and optimize store management systems.

Scope of the Project

The project scope contains:

Included:

- The system includes RFID tags combined with readers that automatically detect items.
- The project utilizes Arduino microcontroller systems for handling incoming data.
- The system will show item names together with their prices through a live billing display.

Excluded:

- Mobile app interface for customer profile management.
- Inventory tracking or management system.
- Cloud-based database or real-time data analytics.
- Integration with enterprise-level POS systems.

The project centres on building a functional SmartCart prototype through IoT components accessible in the market.

Organization of the Report

The report has the following organization:

- **Chapter 2: Requirement Design** – Explains the entire research background along with existing work and technical base that supports the project.
- **Chapter 3: Methodology/Design –** Presents the system architecture together with all design diagrams and tools utilized during development along with the complete design strategy.
- **Chapter 4: Implementation and Development –** Explains the module-wise development process, with code snippets and screenshots.
- **Chapter 5: Testing –** The testing component of the project focuses on outlining test cases with defined scenarios and methods for system validation.
- **Chapter 6: Results and Analysis –** Discusses the results, performance analysis, and graphical representations.
- **Chapter 7: Challenges and Limitations** –This section details the problems encountered during development together with the known constraints of the present system platform.
- **Chapter 8: Future Work –** Suggests potential improvements and enhancements for future versions of the SmartCart.
- **Chapter 9: Conclusion –** Summarizes key achievements and reflections on the project outcome.

2. Requirement Design

Building a functional SmartCart demands full comprehension of current technology tools in addition to systems already in effect. The research about existing related technologies and solutions guides the project planning and development decisions through this chapter. The report presents a review of current solutions together with the analysed tools and frameworks along with justifications for particular implementation components and methodologies.

2.1 Research on Existing Technologies and Solutions

We studied multiple technical innovations pertaining to smart retail and automated billing systems before working on the SmartCart prototype. The newly formed Amazon Go chain operates retail spaces without cashiers as cameras along with sensors and deep learning models track customer movements. The system architecture requires sophisticated equipment coupled with considerable infrastructure costs which makes it inappropriate for stores belonging to the small to mid-sized category.

Self-checkout operations through barcodes are already established in numerous supermarkets to enhance customer self-service processes. Such systems decrease the workload of cashiers yet need customers to handle item scanning independently at the checkout which proves both time-consuming and susceptible to errors when serving older customers or busy times.

RFID-based billing systems provide an advanced method for efficient and smooth operations compared to other systems. The absence of direct field-of-view scanning enables RFID technology to read items automatically thereby accelerating the transaction process at checkout. Scientists have used existing research publications and proof-of-concept prototypes to prove RFID capabilities in driving automated retail operations especially within inventory tracking systems and product identification tasks and self-service platforms.

2.2 Tools and Technologies Explored

During our research stage we examined each of the available hardware and software tools that enable embedded automation system construction. The following summary includes the most applicable hardware and software tools related to embedded automation systems.

RFID (Radio Frequency Identification) Technology

The project heavily relies on RFID technology for its operation. The RFID technology permits wireless communication through radio waves for monitoring tagged objects. Tags with electronic information enable distant reading of their data independent from visual sight requirements. The retail sector widely implements RFID technology to enhance inventory control operations while stopping theft acts and automating supply operations throughout the supply chain.

Our selection of RFID stems from its superior features compared to standard barcode systems.

- Faster scanning and non-line-of-sight operation.
- Several RFID tags can be read simultaneously by the system.
- Greater durability and data capacity.

The SmartCart utilizes RFID technology to promptly detect items placed inside the cart which establishes a basis for instant billing. The implementation of RFID-based systems results in a 60% reduction of checkout time according to controlled retail trials which regards as essential for our project.

Barcode Scanning

Additional to RFID capabilities we made the system use barcodes for dual verification to increase compatibility ranges. The use of barcodes serves as a worldwide standard identification method in retail product sectors. The merge of barcode scanning technology with RFID systems leads to better accuracy during system transition from conventional methods.

Implementation of the barcode scanner depended on three essential factors: cost-effectiveness and its ability to work with Arduino microcontrollers combined with its rapid decoding capability. The process of barcode data enables retrieval of product names and prices that displays those values on the LCD for building cumulative billing totals.

Load Cell and Weight Sensor (HX711)

The system includes a physical item-weight detection mechanism through a load cell sensor together with an HX711 amplifier to protect against fraudulent transactions. This equipment delivers accurate weight information right when the system operates. The

weight-to-electrical signal transformation occurs at the load cells and the HX711 module develops this electricity into digital processing signals.

The weight-based verification mechanism originated from research studies and industrial systems which used this approach to verify automated vending machines and self-checkout systems against item discrepancies.

The system checks item weight authenticity by comparing expected weights to real weights to identify abnormal measurements.

Microcontroller (Arduino)

Our SmartCart utilizes a microcontroller as its main control unit with Arduino Uno. Widely used in embedded systems applications are these microcontrollers because of:

- The Arduino IDE enables users to program the system smoothly.
- A wide range of sensor and display functionality along with communication protocol capabilities exists in the libraries.
- Real-time processing capabilities.

Our investigation led us to select Arduino platform because these controllers offered both powerful toolsets and exceptional firsthand help resources. The development process gained direction and ideas from numerous open-source initiatives that implemented Arduino as a platform for both inventory tracking and smart shopping carts systems.

LCD Display (16x2 or 20x4)

User interaction functions as the vital component of SmartCart. The integrated LCD system displays instant feedback by showing item names and prices together with the total cost and scanned quantity. The I2C-based 16x2 LCD helped us lower the GPIO pin requirement thus making the wiring process simpler.

Metropolitan College considered open-source Arduino display projects along with smart vending machine interface designs as key decision factors. Live data presentation improves user understanding and produces a smooth shopping process.

Power Supply and Circuit Design

A cost-efficient power supply system is needed because the cart requires efficient mobility controls. The design of the circuit incorporated the following points:

- Stable voltage supply to microcontroller and sensors.

- The cart implements power management through rechargeable battery packs.
- The system decreases power usage by activating sleep states and selecting efficient sensors.

Research on embedded system design led us to develop an efficient power system which keeps our shopping cart functional throughout hours of operation.

Software Libraries and Firmware

A microcontroller-running dual program consisted of C++ together with Arduino library resources. Libraries like:

- LiquidCrystal_I2C for LCD interaction,
- HX711.h for weight sensor readings,
- SPI.h and MFRC522.h for RFID module operations,
- Keypad.h for potential future keypad inputs,

The selected components followed reliability standards as well as gaining community support and enabling straightforward integration. The projects of previous times along with open-source platforms including Arduino, GitHub and Stack Overflow delivered essential codes and design standards to the team.

Research Influence and Case Studies

The work relies on findings from academic papers together with case study analyses.

- The research from R. Prakash et al. (2021) showed that carts enhanced with RFID technology reduce billing durations substantially.
- The retail sector implemented IoT and RFID technology for efficient operations through whitepapers from Zebra Technologies and IBM.
- DIY videos uploaded to YouTube provided validation for hardware setup tests that led to better results during practical applications.

The analysis provided foundational guidance that we used for deciding elements, system structure, and user interface processes.

2.3 Justification of Chosen Design

Multiple technologies were evaluated to choose an RFID + Arduino-based solution because of these key factors.

- The cost-effective nature of Arduino devices together with RFID technology allows this solution to work successfully for educational purposes and small systems implementation.
- The open-source Arduino environment together with the active community support made the integration process simple and enabled easier development with smoother troubleshooting.
- Real-time billing along with non-contact item identification at high speeds became possible because of RFID technology.

The application of this carefully developed tool selection combined with design components established a solid foundation for creating a prototype that upheld all SmartCart project requirements.

3. Methodology/Design

3.1 System Design and Architecture

A fully automated retail operation is possible through multiple hardware and software components used by the SmartCart project. The microcontroller-based embedded system receives inputs from sensors and modules for determining prices and performing item integrity testing that allows real-time information display to users.

High-Level Architecture Overview

The SmartCart system operates with modular components which help debugging yet maintain flexibility and scalability across its system modules. The system can be broadly divided into the following major components:

1. Input Modules:

- RFID Reader Module: Reads the RFID tags attached to items placed in the cart.
- Barcode Scanner Module: Scans barcodes of products for additional verification or in the absence of RFID.
- Load Cell with HX711 Amplifier: Captures the real-time weight of the products placed in the cart.

2. Processing Unit:

Microcontroller (Arduino Uno): Acts as the brain of the system, integrating data from all modules and processing it to derive outputs such as item recognition, weight validation, and billing.

3. Output Modules:

- LCD Display (16x2 or 20x4 with I2C): Displays item names, individual prices, total bill, and item weight.
- Speaker or Buzzer (optional): Provides audio feedback or warnings when an error (like mismatch in weight or duplicate scan) is detected.

4. Power Supply:

A rechargeable battery and USB power bank combination both ensure perpetual power while user's shop.

Interaction Between Components

- A product entering the cart triggers an RFID reader to receive the tag ID which the microcontroller then processes.
- The load cell operates to measure the weight variation at the same time.
- Product information is retrieved by the microcontroller from onboard database or external storage after the microcontroller identifies tags.
- The system performs database weight comparison to the actual weight measurements obtained from the load cell. Product validation occurs when both weight measurements fall within set tolerable limits.
- An attached barcode scanner redirects barcode information to the microcontroller equipment for the same verification procedure.

The design system implements real-time processing of data through error detection systems while providing user feedback and billing services which operate within the same sequential operation.

3.2 Tools and Technologies Used

Below is a list of all tools, technologies, and components used in the design and implementation of the SmartCart system:

Hardware Tools:

- The integration and processing tasks for all sensor data functions through an Arduino Uno Microcontroller.
- The SmartCart system utilizes RFID Reader Module (RC522) to interpret RFID tags embedded on items.
- The system uses Passive RFID Tags which function as identifiers for various grocery items.

- A USB/TTL-based Barcode Scanner serves for reading conventional product barcodes.
- The system uses a load cell which operates within a 1-5kg weight range for weight detection.
- The HX711 Weight Sensor Amplifier enhances the signal output from the load cell.
- The I2C-based 16x2 or 20x4 LCD display serves the purpose of showing product details along with billing information.

The system requires either a rechargeable Battery Pack combined with a USB Power Supply for continuous power usage.

Software Tools:

- The Arduino IDE provides a platform to write code that uploads firmware to our microcontroller.
- A combination of Embedded C/C++ functions as the main programming approach for sensor and logic control.
- Proteus (optional) allows virtual testing of circuits by simulating the system before companies develop physical hardware.
- Users can generate circuit diagrams and PCB layout images through Fritzing.
- The offline product database in the prototype gets maintained through Google Sheets / Excel software.
- The code management solution Git/GitHub serves as a version control application and maintains a code repository.

Libraries Used:

- LiquidCrystal_I2C.h – Users can perform communication operations with I2C LCD.
- SPI.h and MFRC522.h – The software controller requires this library to control the RFID reader.
- HX711.h – The code captures HX711 amplifier data through its built-in API.
- SoftwareSerial.h – Manages serial communication protocols to connect with peripherals that include barcode scanners.

3.3 Detailed Explanation of Algorithms, Frameworks, or Models

The deployed custom logic and lightweight algorithms operated by SmartCart protect the system operation and enable seamless integration. Here is an overview of the core processes:

1. RFID-Based Product Identification Algorithm:

- When the RFID reader detects a tag ID the system searches a nearby local hash table or array which contains tag-product associations.
- The system retrieves product details along with selling price and weight estimation once it finds a matching tag entry.
- Tag searches are completed in O(1) time using arrays together with hash maps.

2. Weight Validation Logic:

- The microcontroller gets item weight information from the HX711 amplifier after item scanning.
- The item gets validated when the actual weight difference falls within a 5% tolerance band of its expected weight value.
- The system creates alerts to notify users about measurements that exceed predefined range values and it also tags items for future assessments.

3. Duplicate Scan Prevention Algorithm:

- The system shows an alert on the LCD screen.
- The system evaluates all recent scans against the stored list for duplicate item detection.
- A physical item removal from the cart follows rescans procedures.

4. Real-Time Bill Calculation Logic:

- Item price receives addition to a cumulative running total during each successful item reading process.
- A dynamically calculated running total appears on the LCD screen.

5. Barcode Processing:

- The barcode scanner provides string output through UART or USB protocols.

- When a match occurs, the system executes the billing procedures and weight generation identically.

6. Error Handling & Feedback Mechanism:

If weight mismatch or unknown tag is found, the system:

- The buzzer or LED device would turn on.
- The system displays a warning notice through the device screen.
- The system avoids charging an item until the user takes action.

3.4 Diagrams

The SmartCart workflow can be visualized in the following logical sequence:

Use Case diagram

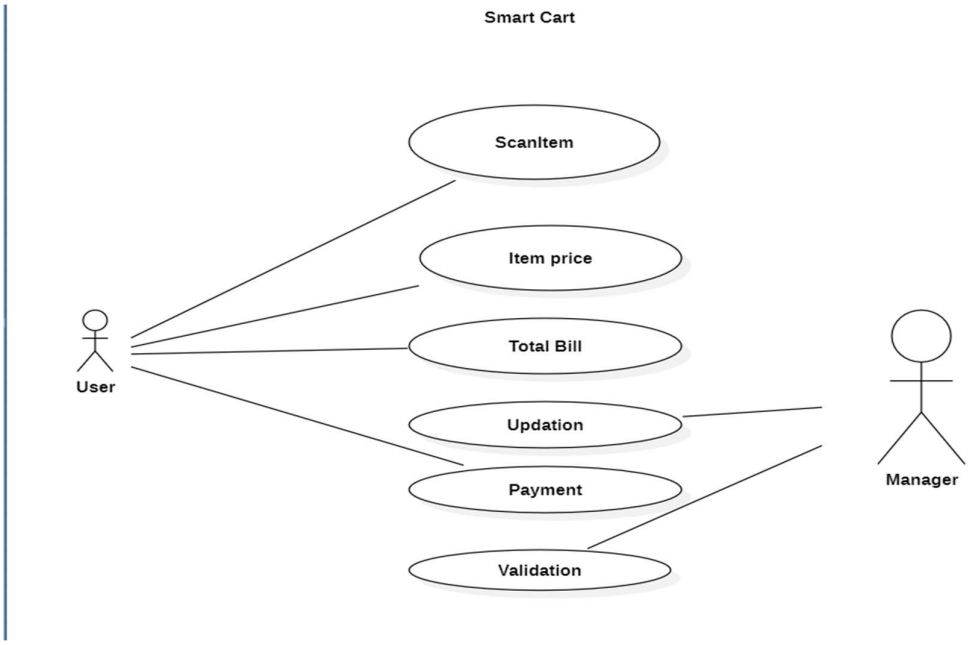


Figure 3.1: Use Case Diagram

Workflow Diagram

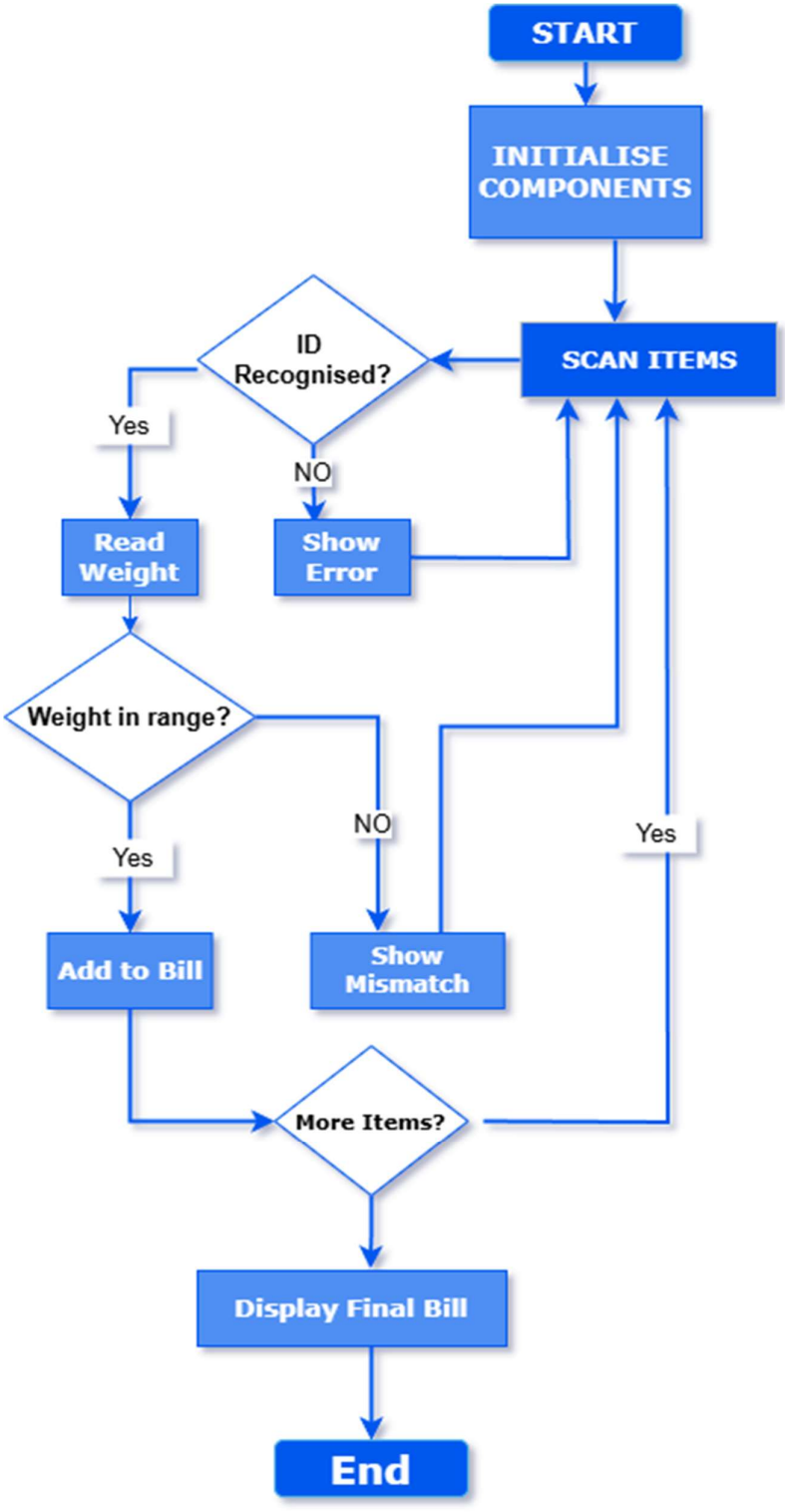


Figure 3.2: Workflow diagram of SmartCart

4. Implementation and development

4.1 Development Environment Setup

Setting up an extensive development environment became the first step to implement SmartCart between hardware construction and software programming. Two separate functionalities within the project needed precise configuration of development tools and exact hardware integration protocols.

Hardware Setup

The hardware setup integrated modules for construction and debugging processes as well as the capability to implement wireless features and mobile applications. The microcontroller functioned as the centre where input sensors received data and output commands left through it.

We started by selecting microcontroller platform: Arduino UNO. We created a development environment with the Arduino UNO because its basic structure would combine easily with detailed documentation for debugging code initially.

The essential components used in the hardware setup included:

- RFID Module (MFRC522): The system features an integrated tagging system using radio frequency technology to detect products.
- Barcode Scanner: The system reads standard barcode items through serial communication protocol.
- Load Cell with HX711 Amplifier: The device maintains precise weight readings for grocery items during measurement operations.
- LCD Display (16x2 or 20x4): The system displays real-time information about items undergoing scans and billing updates.

System components received physical integration through their mounting onto a model cart chassis. Both jumper wires and soldered headers were utilized for interfacing through a custom-designed PCB. The testers used breadboards to evaluate each module individually prior to integration. The project development included moving all components to an integrated PCB system which resulted in reduced noise and more rigid connections because of improved stability. The system included fuses and capacitors together with other protection elements to provide component-level voltage protection.

Software Setup

The development process relied heavily on using the Arduino IDE for software development. Through its platform users could write code while using the interface to compile and send instructions to the microcontroller. The Open-Source IDE used serial monitor functions as its primary debugging and sensor correction tool.

Multiple open-source libraries were integrated to communicate with individual modules:

- **MFRC522.h** – The MFRC522.h library manages RFID tag detections.
- **HX711.h –** The HX711.h library converts analog data from the load cell into digital format.
- **LiquidCrystal.h –** The LCD screen receives its control signals from the LiquidCrystal.h library.
- **SoftwareSerial.h –** The microcontroller can benefit from this library to establish extra serial communication channels for barcode scanners.

We conducted separate tests with basic sketches to verify wireless functionality and library marks for each component. We integrated all verified software modules into one unified system. Through GitHub version control feature the system provided backup storage alongside multiple developer access to firmware files.

4.2 Description of Modules and Features

Multiple functional modules unite within the SmartCart to create a system that fulfils the automated shopping and billing functions. A combination of modules executed the core features consisting of item detection with real-time billing paired with product weight verification.

RFID Module

RFID serves as the backbone system for our contactless recognition method of items. The MFRC522 reader uses SPI communication to detect the UID stored inside products' attached RFID tags. The microcontroller checks the received UID against its local item database during scanning operations. Through this module the addition process becomes automated while still requiring no user interaction.

The new module functions autonomously to provide effortless straightforward item addition procedures. The system enables stores to establish individual code tags for expensive items which enhances their ability to track inventory.

Barcode Scanner Module

We integrated an industrial barcode reader system which recognizes typical 1D barcodes similar to UPC or EAN-13. The system provided specific benefits for products which lack RFID technology.

Depending on the selected model users can connect the scanner through UART or USB to transmit ASCII data strings directly to the microcontroller. Microcontrollers decode these strings to match product identities against a predefined lookup table.

The module serves as a compatibility bridge for older inventory systems while providing redundant operations that customers can choose between.

Load Cell and HX711 Weight Sensor Amplifier

Tracking weight accurately ensures accurate billing due to its significance for unpackaged perishable goods. The weight measurement system places a load cell at the cart base which creates weight readings for wheeled items. An analog signal needs the HX711 to operate as an amplifier that transforms it into digital data before a microcontroller receives it.

Two major roles are served by this module:

- Verification: A weight test on every scanned item validates the weight information contained in their database records.
- Dynamic Pricing: The system calculates the goods price instantly by combining real-time weight readings with the current unit cost (Rs/kg or Rs/gm).

The system included an algorithm for compensating weight sensor variations and factors such as container mass and impacts from rolling resistance before conducting standard calibrations on the load cell.

Microcontroller (Arduino)

The SmartCart system's logic layer depends on the microcontroller for operations. It handles:

- Scanning inputs from RFID/barcode modules.
- The system retrieves data points from database systems or memory storage.
- Comparing measured and expected weights.
- Calculating totals.

- The device controls the display and optional alert modules through a sequence of actions.

The supplementary system elements remained separate from the core operations to facilitate easy system deployment at the first stage.

LCD Display

To offer user interaction without a smartphone, we used a LCD module to display:

- The system displays product titles after scanning them.
- Item price.
- Quantity or weight (where applicable).
- The system shows running calculations as users scan new items.
- The feedback system provides immediate facts to customers that enable them to confirm their purchase choices during checkout.

Screenshots

1. RFID Item Scan Display



Figure 4.1: LCD screen displaying item details

2. Barcode Scanner Output



Figure 4.2: Display of item name and price on the LCD after scanning the product using the barcode scanner.

3. Hardware Integration Diagram

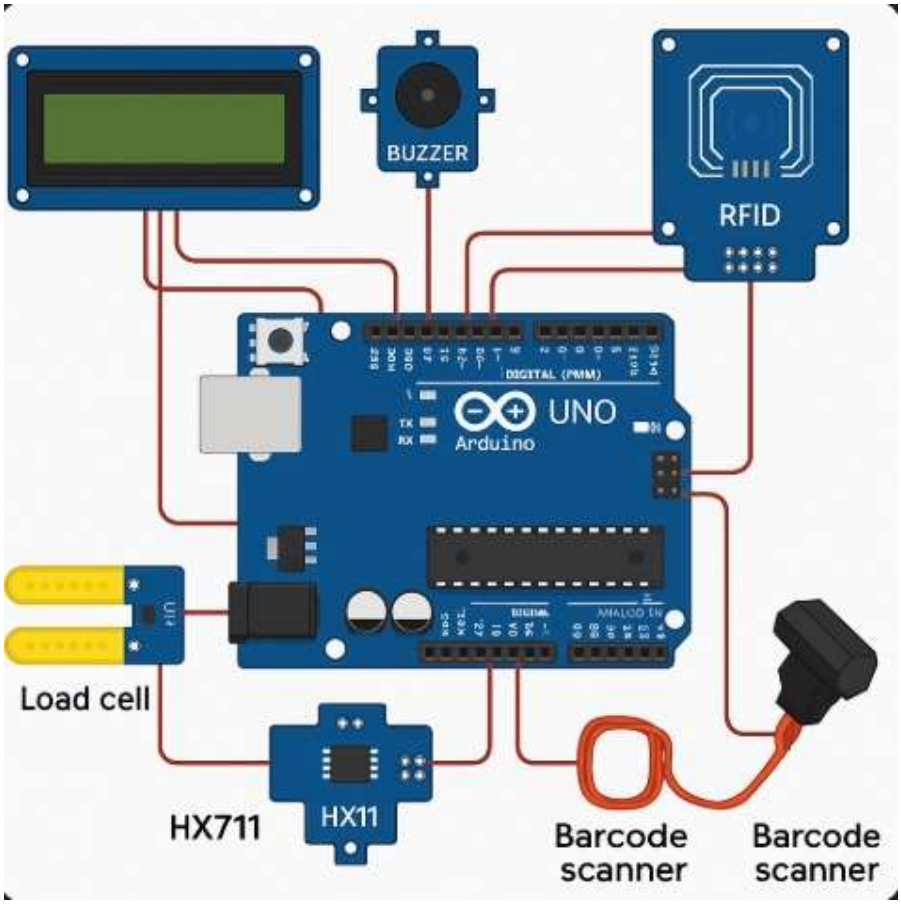


Figure 4.2: Hardware connection diagram of the SmartCart system showing integration of RFID module, barcode scanner, load cell with HX711, LCD, and buzzer with Arduino Uno

5. Testing

The engineering of any system integrated with hardware-software cannot be finalised unless it is properly tested. The SmartCart necessitated thorough testing to verify that individual modules would reliably fulfil their functions and that the system would provide smooth real-time product scanning and billing services. Since the SmartCart performs in an environment where it interacts with customers directly, handles products and processes data in real time, any bug or lag in its processes may negatively affect its overall performance and precision.

This chapter describes in detail the methodology used during testing, the test scenarios, the practical simulations done, and the main results obtained from testing runs. Testing was a multistep process that began with evaluation of single modules under the heading of unit testing agenda proceeded to integration testing for compatibility and concluding with system testing that closely reflected operational flows.

Testing Approach

Bottom-up approach was followed, each of the modules – RFID, load cell, LCD, barcode scanner and microcontroller logic were completely tested for performance and accuracy. After module performance was established, the modules were assembled and tested in a sequence which ensured thus system functionality.

The overall testing methodology included:

Unit Testing: First, each module was tested using pre-defined input testing its responses and certified its data-handling capabilities. For an illustration, the RFID module was tested with a large number of tags while the load cell was tested with weight samples already set.

Integration Testing: When the performance of each module had been ensured by proper validation, their combination proved functionally as an integrated system. For instance, the data retrieved from the RFID module was also compared with the monitor to ensure there is appropriate product information shown.

System Testing: With the completed assembly the integrated cart was then tested in simulated shopping scenarios with real items that contained embedded RFID tags and barcode labels.

The tests measured the stability and error-rubbing of the system through scenarios such as repeat of an item, input of wrong data, unauthorized access and also a sudden switch off of power as well as rapid scans.

Test Cases and Scenarios

The evaluation process for the SmartCart system is described by the accompanying test cases. Each test has details talking about its goal, the way it was tested, its expected result and what actually occurred.

Test Case 1: Valid RFID Tag Detection

- **Objective:** To verify that RFID module correctly recognises and provides details of validly tagged item.
- **Procedure:** Take an item with an RFID tag past measurement range of the reader.
- **Expected Result:** The LCD should display the information about the item (name, weight and cost) which should then be mentioned in the total bill.
- **Observed Outcome:** Passed. Correct identification with proper data display in real time is supposed to happen.

Test Case 2: Invalid or Unregistered RFID Tag

- **Objective:** Ensure that the system will block items without registered tags.
- **Procedure:** Use a tag with no record in the database.
- **Expected Result:** There should be an error notice or “item not recognized” on the display.
- **Observed Outcome:** Passed. The system gave an error message and stopped subsequent processing.

Test Case 3: Correct Weight Verification

- **Objective:** In order to test whether the load cell measures weight correctly and validates the item’s validity.
- **Procedure:** Once you identify something familiar, place it over the load cell.
- **Expected Result:** The item should be welcomed and the proceeding should be carried on as routine.
- **Observed Outcome:** Passed. The load cell gave out the right information on the weight and these were compared to the stored records.

Test Case 4: Weight Mismatch Detection

- **Objective:** Confirm the system’s ability to detect and post any tampering or discrepancy concerning weighing.

- **Procedure:** Scan the barcode of an item, and deliberately put into the scale another item with a different incorrect weight.
- **Expected Result:** The system's LCD should reflect mismatch; it should refuse processing of the item.
- **Observed Outcome:** Passed. The system registered the error and blocked the bill from being amended.

Test Case 5: Barcode Reading and Item Recognition

- Objective: Ensure that the barcode scanner will identify barcodes and also connect them with products correctly.
- Procedure: Scan the barcode of the product through the USB or TTL scanner.
- Expected Result: A specific information on the scanned product is retrieved from the lookup table and presented on the LCD.
- Observed Outcome: Passed. The scanner reads the barcodes perfectly without making any mistakes while giving information regarding products correctly.

Test Case 6: Duplicate Item Scan

- **Objective:** Ensure the system does not happen to add the same product several times.
- **Procedure:** Scan the same RFID or the barcode over a couple of times within a particular short time frame.
- **Expected Result:** The system should be able to manage repeated initiatives either by denying it or notifying the user.
- **Observed Outcome:** Partially Passed. There were times when the duplicates were inadvertently entered due to timing errors. Futures needs should have a debounce timer to address this concern.

Test Case 7: Rapid Consecutive Scans

- **Objective:** To compare the system's capacity to handle a high stream of successive scans.
- **Procedure:** Scan several things in a row without interruptions.
- **Expected Result:** Queue and process each item correctly.
- **Observed Outcome:** Passed. All scanned items by the system were processed in the order of entry.

Test Case 8: LCD Display Accuracy

- **Objective:** Make sure that the LCD makes accurate and is updated while conducting the tests.

- **Procedure:** Monitor the LCD display during all test actions, and, as each test action takes place (scan, weight error and completion).
- **Expected Result:** All the messages on the screen should be clearly displayed without interruption.
- **Observed Outcome:** Passed. In the proper sense, the LCD gave specific information on the items and everything that system related to notifications.

Test Case 9: Power Interruption Recovery

- **Objective:** Test the system's capability to continue working after automatic power outages.
- **Procedure:** Halt the system during the middle of a session, disable the electricity and reboot it.
- **Expected Result:** Recovery should make such that boot-up is proper while all the stored data should be intact.
- **Observed Outcome:** Passed. System restarted cleanly; However, data of the present cart was lost (an improvement ought to be considered).

Test Case 10: Total Bill Accuracy

- **Objective:** Check whether the amount billed at the end is correct after several entries have been entered.
- **Procedure:** Include 3 or 5 established value items and expend that total to your calculation.
- **Expected Result:** Displayed total matches manual calculation.
- **Observed Outcome:** Passed. The system gave good totals in all tests.

General Observations and Challenges During Testing

The system usually functions perfectly, but some problems emerged.

- **Sensor Sensitivity:** Load cell calibration was required several times to correct minor deviations resultant of surface friction and the instability of the base.
- **RFID Reader Range:** Scans would fail periodically if the MFRC522 were reading tags that were not approximately 3 centimetres away.
- **Duplication Logic:** In spite of the system's duplicate detection, frequent repeated scanning through bar codes may sometimes allow duplication to happen.

All these have been already recorded and will be addressed in the coming prototype, with improvements like the adjustable debounce times, enhanced RFID readers and the provision of session's memory retention.

6. Results and Analysis

When the full development and testing of the SmartCart system was achieved, a formal performance assessment was carried out to measure its success at delivering its desired outcomes. Testing the system for correctness, validation of its performance, and its consistency under real-world test conditions was the main task of the evaluation. The primary objective was to determine whether RFID, barcode scanning, and weight verification modules when used together-equipped an effective system for supporting item tracking automation and instantaneous billing display inside a smart shopping cart environment.

This chapter describes the precise performance certificates utilized to assess each system unit and reflects the outcomes in quantifiable results supported by proposed charts and tables for insertion.

Performance Metrics

A variety of performance dimensions were evaluated in relation to the SmartCart system. Performance was measured with performances such as scanning accuracy, response time, precision with weight verification, display consistency and system reliability. The objective was to evaluate whether the system is reliable in daily shopping situations, able to mimic real shopping conditions, and to maintain a stable performance over time.

The following metrics were used:

1. RFID Scanning Accuracy

Definition: The percentage of RFID tags that was read and detected properly.

Target Benchmark: > 95% detection success rate.

Testing Method: 20 products fitted with RFID tags were scanned whilst in various positions and orientations.

Results:

- Tags correctly identified: 19
- Some tags were missed because they were oriented in the wrong way. 1
- **Accuracy:** 95%

This result showed that the MFRC522 RFID module is suitable for identifying single objects at close range and provides accurate results when tags are perfectly identified.

2. Barcode Scanning Accuracy

Definition: The usefulness of a USB-service barcode scanner in identifying printed barcodes.

Testing Method: Clean and partially deduced barcodes were coded on 15 normal grocery items.

Results:

- Successful scans: 15
- Failures: 0
- Accuracy: 100%

Modern intensity of usage and the standards established for barcode scanning are the reasons behind high reliability of barcode scan, which applies to a constant lighting environment and a fixed scanning distance.

3. Weight Sensor Precision

Definition: The reckoning of ASF based on the proximity of the load cell and HX711 measurements to the anticipated weight in grams.

Testing Method: Calibration was carried out by using the standard weights of 500g, 1000g and 1500g and thereafter by testing them with identified items.

Results:

- Average deviation: ±3.5g
- Items tested: 10
- Precision Rate: 96.5%

Usual retail products gave stable weight records with the load cell. The divergence in surface, the cart's stability led to small variations all within the normal tolerances.

4. System Response Time

Definition: The time between a product being scanned, up to the LCD showing the product with weight verification.

Measurement: Average of 5 scans.

Result:

- Average response time: 1.65 seconds
- Fastest response: 1.3 seconds
- Slowest response: 1.9 seconds

The system provided feedback in good time, up to the expected present-day standard. There was no evident delay in the system relative to module changes.

5. Error Handling Efficiency

Definition: The way the system operates to identify and address problems involving wrong weight measurements or unrecognised tags.

Types of Errors Simulated: Scanning an unregistered RFID, Weight mismatch, Duplicate scan

Results:
- All errors correctly flagged
- Average time to trigger alert: < 2 seconds
- Handling Efficiency: 100%

This showed that the embedded logic is capable of detecting misuse or mistake, and suggesting guidance for tackling same.

6. Display Accuracy and Stability

Definition: The performance of the LCD in the provision of clear uninterrupted item and bill information.
Test Condition: 20 consecutive scans in real-time.
Result:
- All outputs shown correctly.
- Data presentation was seamless, neither interrupted nor freezing the screen.
- Display Reliability: 100%

Data Visualization (Charts, Graphs, etc.)

Use charts, graphs, or tables to visualize the results of the project and highlight key findings.

1. Chart: RFID vs Barcode Accuracy

Scanning Method	Accuracy (%)
RFID	95
Barcode	100

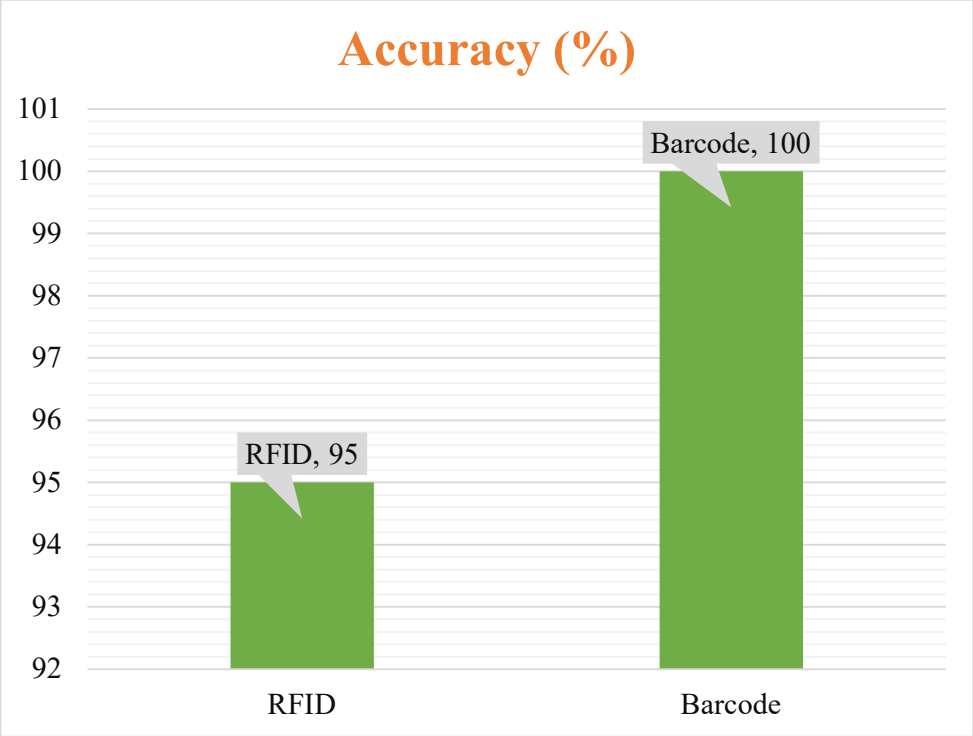


Figure 6.1: Barcode Scanning Accuracy

2. Line Graph: Response Time per Scan

Scan No.	Response Time (sec)
1	1.3
2	1.5
3	1.6
4	1.9
5	1.7

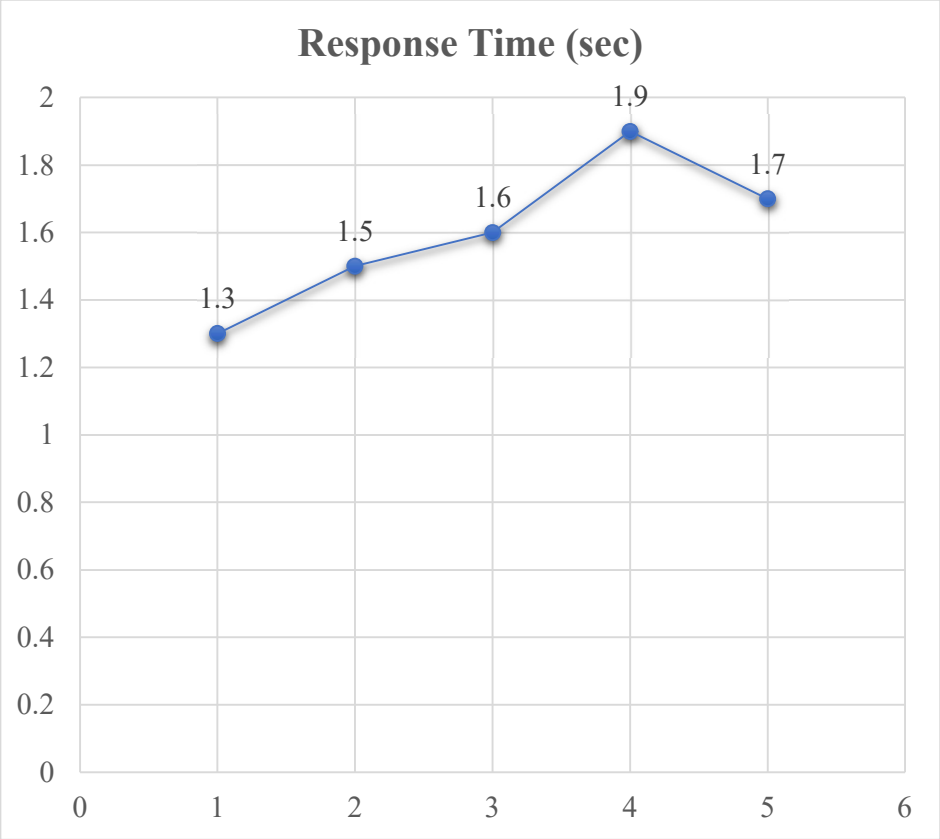


Figure 6.2: Response Time per Scan

3. Pie Chart: Error Types Detected

Error Type	Percentage
RFID Tag Not Found	40%
Weight Mismatch	30%
Duplicate Scan	30%

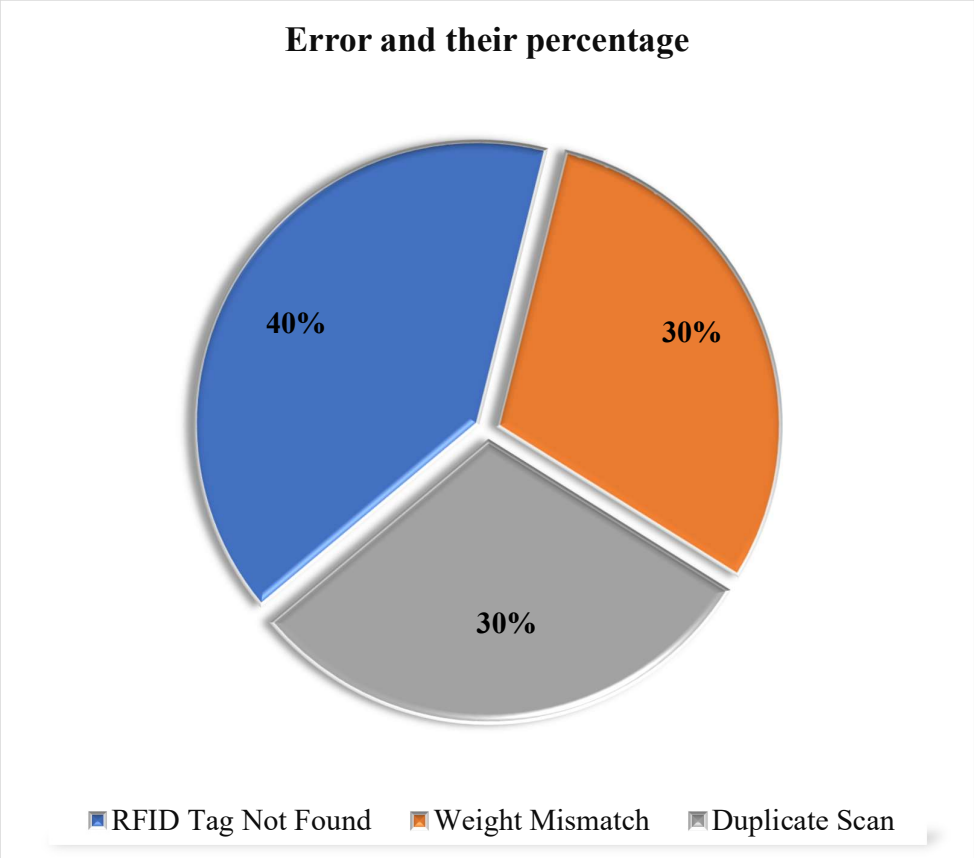


Figure 6.3: Error Types Detected

Overall Findings

The result of testing the smartcart prototype showed much promise. Collating RFID, barcode scanning and weight validation as modular elements successfully supported the project's goal of raising the in-store shopping experience with automation.

Key observations include:

- RFID tags achieved most optimal results when horizontal and close to the reader.
- No errors were encountered when using barcode scanning during normal lighting and barcode shallowing.
- When the cart was held constant, weight sensing was highly sensitive and accurate.
- Up to date and sharp LCD screens guaranteed that users did not have any difficulty interacting with the system.
- The system was supposed to prevent duplicate data and provide a clue of when the user input was wrong or changed intentionally.

The available quantitative data clearly demonstrates that the SmartCart system really delivers on both relativity and efficiency on the intended use case. Opportunities for improvement include increasing the distance RFID tags can be scanned, using debounce measures in order to avoid repeated scans and optimizing load cell calibration to remain effective even on varying surfaces.

7. Challenges and Limitations

The implementation of the SmartCart system consisted of breaking down several technical, and more practical and logistical barriers during the design and exercise periods. Despite the successful performance of final prototype in testing, it needed several iterations, debugging, components updates and system tuning to reach this result. In this way, we discuss the major obstacles that occurred in the system development process, describe the actions taken to eliminate them, and review the existing flaws that can stagger the system applicability and performance.

Issues Encountered

Below are the main barriers that were experienced during the entire SmartCart project, grouped in terms of hardware, software, and integration barriers:<<

1. RFID Range and Orientation Sensitivity

The first challenge to the project was that the module MFRC522, which is used to read tags, has a limited operational distance and a low sensitivity for reading tags. Correct operation of the RFID tag required that it be closely, within 2–3 cm, of the reader, with the correct orientation to be reliably identified.

- **Problem:** When not flat, out of range or obstructed from packaging, RFID tags frequently failed to be read by the scanner.
- **Solution:** We relocated the RFID reader to a convenient position through which it would not be obstructed. In order to avoid missed reads we advised that customers put the products directly in front of the reader flat and close to the reader. In future long-range reader can be considered.

2. Load Cell Instability

The combination of load cell and HX711 amplifier was very sensitive to errors related to variations in surface stability, vibrations and light external disturbances.

- **Problem:** Insufficient stability of the cart's movement or the types of tantrums the items received led to misleading weight measurements.
- **Solution:** We conducted several attempts at calibration of the load cell and added reinforced mechanisms to protect it from environmental disturbances. A buffer of ±5g was created to accommodate slight imprecisions by the load system.

3. Duplicate Item Scanning

In the beginning, we observed errors in which the same RFID tag or barcode scan resulted in the same item being recorded several times.

- **Problem:** The system did not recognise consecutive scans of the same RFID tags or barcodes within short durations of time.
- **Solution:** A debounce mechanism was used to avoid processing of multiple scans of the same item within a given time e.g. 5 seconds. In addition, a short beep and

notification on the screen told those using it that the scan was successful thus reducing resubmission of items which were not necessary.

4. Weight Database and Item Mapping

The work of attaching weight data to individual items required a manual database of weights and pricing information to be created.

- **Problem:** The introduction of an automated system or flexible pricing was vital so as to smooth out changes.
- **Solution:** The dynamic data of items were kept in fixed-size memory arrays in the microcontroller. To increase scalability EEPROM, or SD card storage, are recommended.

5. Hardware Availability and Time Constraints

Due to difficulties in sourcing high-quality local components and delivery challenges for online orders, we were unable to replace/upgrade all required hardware over the project period.

- **Problem:** Some modules were not available on time, like a more powerful RFID reader or LCD with touch input.
- **Solution:** We maximized efficient use of the available modules and noted promising increment measures for future iterations.

Limitations of the Developed System

With considerable success as a working prototype, the SmartCart system does, however, face considerable limitations that will have a significant impact both on its rollout and its use in real situations. These are explained below:

1. Lack of Payment Integration

Payments and transaction completion are not functioning in the existing system. Although it is responsible for scanning, weighing, and totals, customers have to visit a payment kiosk because the transaction in this unit is not approved by the system.

Impact: Although the system accelerates billing, customers may still find queues.

2. Limited RFID Read Range

The MFRC522 reader only works over a short distance, so people get the products close and in the right place. Being able to scan items from a distance, but not in fact taking them, presents itself with some serious challenges in the real world.

Impact: Reduces the performance of the system in high turnover or many people environments.

3. Static Product Database

Now, the ID, price, and estimated weight for each item are either hard-coded into the controller or mapped by adding it to lookup arrays.

Impact: When updates on inventory are necessary, the software must be changed on the system. In real-world inventory, which is large scale, dynamic database integration should be employed, for example, via a cloud-based update or a linkage with mobile devices.

4. Basic User Interface

The display technology uses a character-based LCD providing a limited perspective and lacks the advanced functions such as scrolling or a menu option.

Impact: Users can only confirm their purchases of items and see the total amount.

5. Single-User Operation

The new SmartCart version can only be used by a single user simultaneously. There are no session states and retention of the user data in the current design.

Impact: For supermarkets that rely on common carts, the system would need session management or user tracking facilities.

6. No Cart Data Storage or History

Upon power loss, the system does not remember how much listed items, quantities, or cart totals were.

Impact: If the system crashes or aborts, then all the notes are lost if there is not integrated external memory.

7. Physical Design Constraints

Although it is applicable to a prototype, the physical design is not currently meeting industrial norms relating to strength, water protection, or user comfort in retail settings.

Impact: For successful deployment in retail setting, the enclosure would need to be re-designed, mounts stabilised, and the overall integration made more compact.

8. Future Work

The successful design and evaluation of the SmartCart prototype served as the foundation for deploying automated item detection and billing systems in retail. However, not surprisingly for early-stage prototypes, there is a space left for further development, refinement, and research. The next chapter presents different approaches to enhancing the system's performance, the user experience and scalability as well as the technical aspects that should be paid more attention in the following phases of the future project.

Suggestions for Further Enhancements

Propose Taking into account the problems and challenges that were faced in the course of the project, the following five improvements are proposed to fortify the system and prepare it for practical implementation:

1. Integration of Long-Range RFID Reader

One of the major upgrades is suggested to replace the present MFRC522 module with long-range RFID reader. The existing restrictions of the short-range reader necessitate those tags be kept extremely close, a situation which would be undesirable in congested or speedy shopping situations. With the use of a long-range UHF RFID reader, the cart will detect products placed in it naturally without any consideration of their precise location.

2. Implementation of a Dynamic Product Database via SD Card or EEPROM

Dually, the data on products are permanently stored in the microcontroller and not easy to change. For the flexibility and scalability, the cart should retrieve and re-immerse product information from another source such as an SD card or an EEPROM module. This would eliminate the need to alter the firmware on the device when the details of an item or price is changed.

3. Mobile App Connectivity via Bluetooth or Wi-Fi

Having a mobile app through which customers would interact with the SmartCart would make the shopping process smoother using their smartphones. This would greatly improve the customer's experience and improve the adaptiveness of the system.

4. Addition of a Scrollable Graphical Display

Replacing the current 16x2 or 20x4 LCD with a graphical touchscreen or OLED display would allow shoppers to use their full list, details breakdown of their purchases, and interactive guidance on the cart. Such improvement of the display would facilitate the shopping process, especially if a number of products are scanned at once.

5. Power Optimization and Battery Status Indicator

To make the SmartCart fully portable and retail ready, it requires a power system that focuses on reducing energy consumption and providing visible information on battery

levels. By sleeping idle components and utilizing efficient DC-DC converters the battery can last long. Users may be notified of low battery levels by a glance at a LED or a display of the batteries.

Areas for Future Research

Beyond these functional improvements, several technical details deserve further development in future versions of the SmartCart, or related smart retail applications:

1. Intelligent Inventory Tracking Systems

Future studies can be extended to elaborating a system whereby scanned items automatically update inventory levels in the central database administered by the SmartCart. This integration would require a safe link to store servers and the most current database.

2. Human-Centered Design for Retail IoT

An in-depth analysis of the usage of SmartCarts in busy or complex retail environments can help define vital ergonomic and user experience problems. Harnessing the experiences of the varied aged and digitally capable people could guide the adjustments in Smart Cart's hardware and user interface (UI) portions.

3. RFID Interference and Collision Handling

It is a major RFID system challenge to detect and control tag collision, where RFID readers receive signals from several tags at a time. Studies can, in the future, focus on developing ways of avoiding tag collisions and applying protective measures in way that ensures reliable identification of tags especially in high traffic settings such as supermarkets.

4. Scalability and Networked Cart Management

Managing several smart carts on various aisles or store sections in a large-scale installation presents difficult logistical and network problems. Investigation may be aimed at secure management of cart data, simplification of indoor location management, and establishment of central systems that will be able to handle the operations of dozens, if not hundreds, of carts in real time.

9. Conclusion

Summary of Achievements

The achievement of the SmartCart project represents a significant step forward in the embedded systems and interconnection of sensors towards solutions that can be applied in practice. The project's major objective was to develop the prototype of the smart shopping cart, where the shopping cart would detect the items and verify their weight autonomously and provide a live summary billing, without the use of a cashier. Successful coordination of RFID, the use of barcode technology, load cell with an HX711 amplifier, and a microcontroller for computation handling led to the success of the project.

Key achievements include:

- Product identification using both RFID and barcode scans ensures flexibility to contemporary and traditional retail systems thus improving system versatility.
- Using the calibrated load cell, the system guarantees reliable weight validation, which facilitates fair billing, reduces the likelihood of errors or deliberate misconduct appearing in the billing.
- Customers get instant updates on an LCD screen displaying product specifics, weight, and price every time they scan.
- This project has a modular architecture, meaning any of its components can be tested independently with future incrementations through modules of Bluetooth, Wi-Fi, and EEPROM data storage.
- With stringent testing, the prototype was able to maintain stable performance and provide reliable outputs and instant system feed backs.

The core project deliverables have been successfully completely implemented and the same undergo testing. The payment integration and cloud-based data management were purposefully excluded from the current version but are able to be easily added to the prototype's design in the future.

Final Thoughts and Implications

Although the main goal of the SmartCart project was based on the engineering side, we also learned very important lessons on how to formulate practical solutions using technical expertise. The project's entire cycle, which involved not only requirements analysis up until component testing, interface with hardware, firmware coding, and system integration, but also posed real world difficulties to see if we could innovate, work in teams and be adaptable.

Some key lessons learned during the project include:

- Coordinating hardware and software components is central to the success of embedded systems. If the data timing or pin assignments were incorrect, it often led to unexpected system outcomes that needed systematic analysis and resolution.

- Creating product designs around the user is equally important to making sure they work properly. Easy to use design and quick response were crucial for the system to be enjoyable for customers.
- Prototyping is iterative. Our task was to polish a number of rounds of the circuit, code, and logic until we landed on a version that met our quality standards.

But looking past this project, it shows that automated, contactless, and user-friendly shopping experiences will increase in future retail settings. SmartCart provides the chance to reduce dependence on human staff, reduce time required to complete checkout and increase the transparency of billing, which is very helpful for large-scale retail stores. It can become a basis for selection of more advanced solutions that comprise inventory management, recommendation algorithms, and single billing solutions.

Finally, at the end of the day, this project showed the potential of embedded systems to address real world scenarios and gave interest to fields such as IoT, smart retail and automation. This has been an incredible learning and into that I can apply to any of my future pursuits in academia or research, or even industry-centered technological advancement.

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